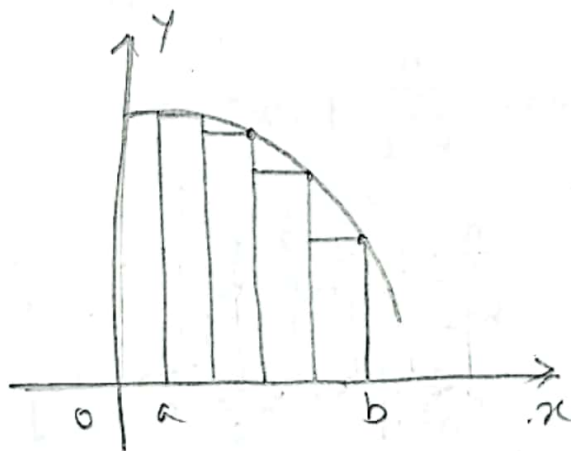
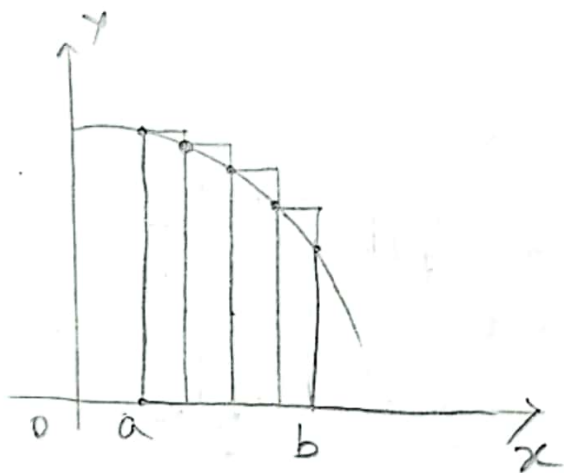


Area under a curve:

If the function f is continuous on $[a, b]$ and if $f(x) \geq 0$ for all x in $[a, b]$, then the area A under the curve $y = f(x)$ over the interval $[a, b]$ is defined by,

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x_k$$



The values of $x_1^*, x_2^*, \dots, x_n^*$ are chosen arbitrarily.

Some common choices being,

- i) the left endpoint of each subinterval.
- ii) the right endpoint of each subinterval.
- iii) the midpoint of each subinterval.

These choices are given by,

$$x_k^* = a + (k-1) \Delta x \longrightarrow \text{Left endpoint}$$

$$x_k^* = a + k \Delta x \longrightarrow \text{right endpoint}$$

$$x_k^* = a + (k - \frac{1}{2}) \Delta x \longrightarrow \text{midpoint}$$

Example: Use x_k^* as the right endpoint of each subinterval to find the area between the graph of $f(x) = x^2$ and the interval $[0, 1]$. [When number of subinterval is not mentioned]

Solⁿ: The length of each subinterval is, $\Delta x = \frac{b-a}{n} = \frac{1-0}{n}$
 $= \frac{1}{n}$

$$\Delta x_k^* = a + k \Delta x = \frac{k}{n}$$

$$\text{Thus, } \sum_{k=1}^n f(x_k^*) \Delta x$$

$$= \sum_{k=1}^n \left(\frac{k}{n}\right)^2 \frac{1}{n} = \frac{1}{n^3} \sum_{k=1}^n k^2$$

$$= \frac{1}{n^3} \left[\frac{n(n+1)(2n+1)}{6} \right]$$

$$= \frac{1}{6} \left(\frac{n}{n} \cdot \frac{n+1}{n} \cdot \frac{2n+1}{n} \right)$$

$$= \frac{1}{6} \left(1 + \frac{1}{n} \right) \left(2 + \frac{1}{n} \right)$$

$$\text{area, } A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{6} \left(1 + \frac{1}{n} \right) \left(2 + \frac{1}{n} \right) \right]$$

$$= \frac{1}{3}$$

Example: Use x_k^* as the left endpoint of each subinterval to find the area under the curve $y = 2x - x^2$ over the interval $[-1, 3]$; $n = 4$.

Solⁿ: The length of each subinterval of $[-1, 3]$ is

$$\Delta x = \frac{3 - (-1)}{4} = 1$$

$$\text{when } k=1, \quad x_1^* = -1 + (1-1) \cdot 1 = -1$$

$$k=2 \quad x_2^* = -1 + (2-1) \cdot 1 = 0$$

$$k=3 \quad x_3^* = -1 + (3-1) \cdot 1 = 1$$

$$k=4 \quad x_4^* = -1 + (4-1) \cdot 1 = 2$$

$$k=5 \quad x_5^* = -1 + (5-1) \cdot 1 = 3$$

$$\begin{aligned} \text{Now, } \sum_{k=1}^4 f(x_k^*) \Delta x &= f(x_1^*) \cdot 1 + f(x_2^*) \cdot 1 + f(x_3^*) \cdot 1 + f(x_4^*) \cdot 1 \\ &= f(-1) + f(0) + f(1) + f(2) \\ &= -3 + 0 + 1 + 0 \\ &= -2 \end{aligned}$$

Example: Use x_k^* as the right endpoint of each subinterval to find the area under the curve $f(x) = 3x + 1$ over the interval $[2, 6]$.

Solⁿ: $\Delta x = \frac{6-2}{4} = 1$

Now,

Right end points of each interval,

$$x_1^* = 2 + 1 \cdot 1 = 3$$

$$x_2^* = 2 + 2 \cdot 1 = 4$$

$$x_3^* = 2 + 3 \cdot 1 = 5$$

$$x_4^* = 2 + 4 \cdot 1 = 6$$

$$\begin{aligned}\text{Now, } \sum_{k=1}^4 f(x_k^*) \Delta x &= f(x_1^*) + f(x_2^*) + f(x_3^*) + f(x_4^*) \\ &= f(3) + f(4) + f(5) + f(6) \\ &= 10 + 13 + 16 + 19 \\ &= 58.\end{aligned}$$

Example: Use x_k^* as the midpoint of each subinterval to find the area between the graph of $f(x) = \cos x$ on the interval $[0, \pi]$.

Solⁿ: $\Delta x = \frac{\pi-0}{4} = \frac{\pi}{4}$

midpoints of each interval,

$$x_1^* = 0 + (1 - \frac{1}{2}) \frac{\pi}{4} = \frac{\pi}{8}$$

$$x_2^* = 0 + (2 - \frac{1}{2}) \frac{\pi}{4} = \frac{3\pi}{8}$$

$$x_3^* = 0 + (3 - \frac{1}{2}) \frac{\pi}{4} = \frac{5\pi}{8}$$

$$x_4^* = 0 + (4 - \frac{1}{2}) \frac{\pi}{4} = \frac{7\pi}{8}$$

$$\text{Now, } \sum_{k=1}^4 f(x_k^*) \Delta x = \left(\cos \frac{\pi}{8} \right) \cdot \left(\frac{\pi}{4} \right) + \left(\cos \frac{3\pi}{8} \right) \cdot \frac{\pi}{4} + \left(\cos \frac{5\pi}{8} \right) \cdot \frac{\pi}{4} + \left(\cos \frac{7\pi}{8} \right) \cdot \frac{\pi}{4}$$

$$= 0.$$

Practice (5.4): 27-30, 35-40, 41-44, 45-48

Examples

Properties of Sums:

$$(a) \sum_{k=1}^n c a_k = c \sum_{k=1}^n a_k, \text{ if } c \text{ is not depend on } k.$$

$$(b) \sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k$$

$$(c) \sum_{k=1}^n (a_k - b_k) = \sum_{k=1}^n a_k - \sum_{k=1}^n b_k$$

Theorem (5.4.2)

$$(a) \sum_{k=1}^n k = 1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

$$(b) \sum_{k=1}^n k^2 = 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$(c) \sum_{k=1}^n k^3 = 1^3 + 2^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Example: Use left endpoints to find the area between the curve $f(x) = x-1$ and the x -axis with interval $[0,1]$.

Solution: The length of each subinterval is,

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{n} = \frac{1}{n}$$

for left endpoint, $\Delta x_k^* = a + (k-1) \frac{1}{n} = \frac{k-1}{n}$

$$\text{Thus, } \sum_{k=1}^n f(x_k^*) \Delta x = \sum_{k=1}^n \left(\frac{k-1}{n} - 1 \right) \frac{1}{n}$$

$$= \sum_{k=1}^n \frac{1}{n^2} (k-1) - \sum_{k=1}^n \frac{1}{n}$$

$$= \frac{1}{n^2} \sum_{k=1}^n k - \frac{1}{n^2} \sum_{k=1}^n 1 - \frac{1}{n} \sum_{k=1}^n 1$$

$$= \frac{1}{n^2} \frac{n(n+1)}{2} - \frac{1}{n^2} \cdot n - \frac{1}{n} \cdot n$$

$$= \frac{1}{2} + \frac{1}{2n} - \frac{1}{n} - 1$$

$$= -\frac{1}{2} + \frac{1}{2n} - \frac{1}{n}$$

$$\text{Area} = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$$

$$= \lim_{n \rightarrow \infty} \left(-\frac{1}{2} + \frac{1}{2n} - \frac{1}{n} \right)$$

$$= -\frac{1}{2}$$

Defⁿ: (Area under a curve)

If the function f is continuous on $[a, b]$ and if $f(x) \geq 0$ for all x in $[a, b]$ then the area under the curve $y = f(x)$ over the interval $[a, b]$ is defined by,

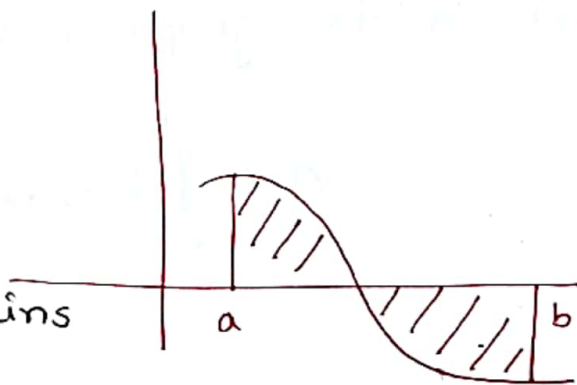
$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x,$$

where Δx is the width of the subinterval and x_k^* is any arbitrary point on k -th subinterval.

This x_k^* can be chosen from the left end point of each subinterval, right endpoint of each subinterval or midpoint of each subinterval.

Net signed area:

If f is continuous and attains



both +ve and -ve values on $[a, b]$ then the limit

A no longer represents the area between the curve $y = f(x)$ and the interval $[a, b]$ on the x -axis, rather it represents a difference of areas i.e the area of the

region that is above the interval $[a, b]$ and below the curve $y = f(x)$ minus the area of the region that is below the interval $[a, b]$ and above the curve $y = f(x)$. We call this the 'net signed area'. So, if the function is continuous on (a, b) then the net signed area A between $y = f(x)$ and interval $[a, b]$ is defined by,

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$$

Here comes the idea of definite integral.

If a function f is continuous on an interval $[a, b]$, then f is integrable on $[a, b]$ and the net signed area between the graph of f and the interval $[a, b]$ is,

$$A = \int_a^b f(x) dx$$

□ $f \rightarrow$ continuous and non-negative $\rightarrow \int f \rightarrow$ area.

$f \rightarrow$ continuous $\rightarrow \int f \rightarrow$ net signed area.

e.g (1) $f(x) = 2x - 2$

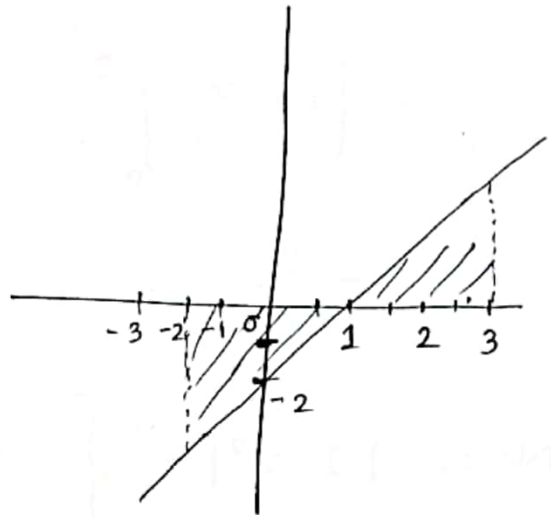
$$\int_{-2}^3 f(x) dx = \int_{-2}^3 (2x - 2) dx$$

$$= [x^2 - 2x]_{-2}^3$$

$$= (9 - 6) - (4 - 4)$$

$$= 3 - 0$$

$$= 3$$



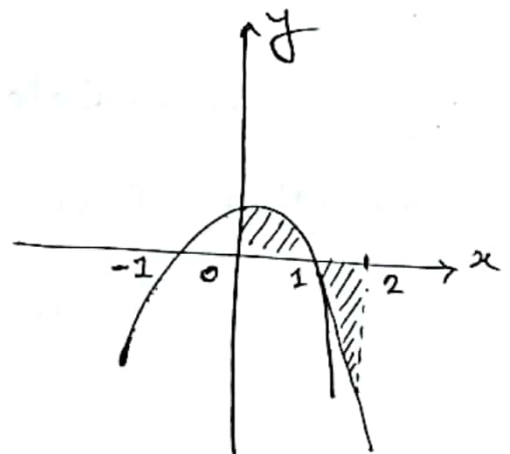
Total area: If f is a continuous function on the interval $[a, b]$ then we define the total area between the curve $y = f(x)$ and the interval $[a, b]$ to be total

$$\text{area} = \int_a^b |f(x)| dx.$$

Example: Find the net signed area and total area between the curve $y = 1 - x^2$ and x -axis over the interval $[0, 2]$.

Solⁿ: Net signed Area

$$= \int_0^2 f(x) dx = \int_0^2 (1 - x^2) dx$$



$$= \left[x - \frac{x^3}{3} \right]_0^2$$

$$= -\frac{2}{3}$$

$$\text{Now, } |1-x^2| = \begin{cases} 1-x^2, & -1 \leq x \leq 1 \\ -(1-x^2), & x < -1 \text{ or } x > 1 \end{cases}$$

$$\text{Total area, } = \int_a^b |f(x)| dx$$

$$= \int_0^2 |1-x^2| dx = \int_0^1 (1-x^2) dx + \int_1^2 -(1-x^2) dx$$

$$= \left[x - \frac{x^3}{3} \right]_0^1 - \left[x - \frac{x^3}{3} \right]_1^2$$

$$= 1 - \frac{1}{3} - 2 + \frac{8}{3} + 1 - \frac{1}{3}$$

$$= \frac{6}{3}$$

$$= 2.$$

Homework: i) Calculate the area under the curve of a function $f(x) = 7 - x^2$ on the interval $[-1, 2]$.

Ans: 18 sq. unit.

(ii) Find the net signed area and total area between the curve $y = -\frac{1}{3}x + 1$ and x -axis over the interval $[0, 4]$.

Ans: $\frac{5}{3}$.

Definite Integral:

Let f be a function defined on a closed interval $[a, b]$. then f is Riemann integrable on $[a, b]$ and the definite integral of f from a to b is denoted by

$$\int_a^b f(x) dx$$

and is defined as, $\int_a^b f(x) dx$

$$= \lim_{\max \Delta x_k \rightarrow 0} \sum_{k=1}^n f(x_k^*) \Delta x_k$$

where x_k^* lies in some subinterval of $[a, b]$ and Δx_k denotes the length of the subintervals. The numbers a and b are called lower limit and upper limit of integration, respectively and $f(x)$ is called the integrand.

Properties of definite integral:

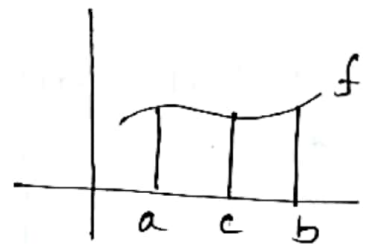
1. $\int_a^a f(x) dx = 0$

2. If f is integrable on $[a, b]$ then,

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

3. If f is integrable on a closed interval $[a, b]$ containing c , then,

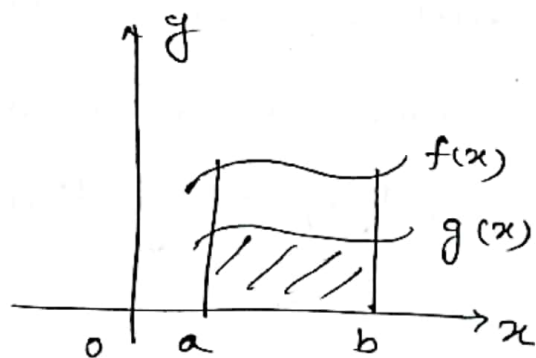
$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$



4. If f is integrable on $[a, b]$ and $f(x) \geq 0$ for all x in $[a, b]$ then $\int_a^b f(x) dx \geq 0$.

5. If f and g are integrable on $[a, b]$ and $f(x) \geq g(x)$ for all x in $[a, b]$ then,

$$\int_a^b f(x) dx \geq \int_a^b g(x) dx$$



□ Fundamental theorem of calculus:

If f is a continuous function on $[a, b]$ then,

1) If $g(x) = \int_a^x f(t) dt$, then $g'(x) = f(x)$.

2) $\int_a^b f(x) dx = F(b) - F(a)$, where F is any antiderivative of f i.e. $F' = f$.

Example: $\int_{-1}^3 \frac{1}{x^2} dx = \left[-\frac{1}{x} \right]_{-1}^3 = -\frac{4}{3}$

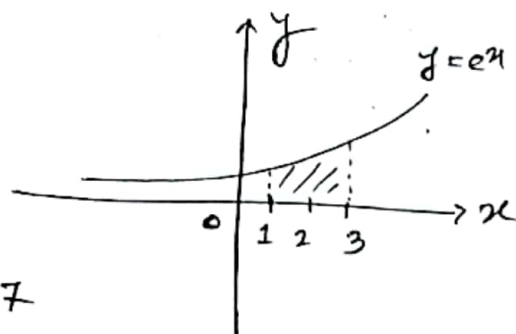
This calculation must be ~~so~~ wrong, because according to the property 5, if $f(x) \geq 0$ then $\int_a^b f(x) dx \geq 0$.

Also the fundamental theorem of calculus applies to continuous function only. It can't be applied

here because $f(x) = \frac{1}{x^2}$ is not continuous on $[-1, 3]$.

Example: Find the area under the curve $y = e^x$ from 1 to 3.

Solⁿ: $A = \int_1^3 e^x dx$
 $= [e^x]_1^3 = e^3 - e^1 \approx 17.367$



Integration by substitution, or u-substitution:

Suppose that F is an antiderivative of f and g is a differentiable function. So we develop a method from the chain rule. We know,

$$\frac{d}{dx} [F(g(x))] = F'(g(x)) g'(x)$$

We can write this in integral form as,

$$\int F'(g(x)) g'(x) dx = F(g(x)) + C$$

or since F is an antiderivative of f ,

$$\int f(g(x)) g'(x) dx = F(g(x)) + C$$

$$\text{Let } u = g(x)$$

$$\Rightarrow \frac{du}{dx} = g'(x)$$

$$\Rightarrow du = g'(x) dx$$

$$\therefore \int f(u) du = F(u) + C$$

This is called the method of u-substitution.

$$\begin{aligned} \text{e.g. (1)} \quad & \int \frac{dx}{\left(\frac{1}{3}x - 8\right)^5} \\ &= \int \frac{3 du}{u^5} \end{aligned}$$

$$\text{Let } \frac{1}{3}x - 8 = u$$

$$\Rightarrow \frac{1}{3} dx = du$$

$$\Rightarrow dx = 3 du$$

$$= 3 \int u^{-5} du$$

$$= 3 \frac{u^{-4}}{-4} + c$$

$$= -\frac{3}{4} \frac{1}{u^4} + c$$

$$= -\frac{3}{4} \left(\frac{1}{3}x - 8 \right)^{-4} + c$$

$$(2) \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$$

$$= \int 2 \sin u du$$

$$= 2(-\cos u) + c$$

$$= -2 \cos \sqrt{x} + c$$

$$\text{Let } \sqrt{x} = u$$

$$\Rightarrow \frac{1}{2\sqrt{x}} dx = du$$

$$\Rightarrow \frac{1}{\sqrt{x}} dx = 2 du$$

$$(3) \int \frac{e^{4/x}}{x^2} dx$$

$$= -\int e^{4u} du$$

$$= -\frac{e^{4u}}{4} + c$$

$$= -\frac{e^{4/x}}{4} + c$$

$$\text{Let } \frac{1}{x} = u$$

$$\Rightarrow -\frac{1}{x^2} dx = du$$

$$\Rightarrow \frac{1}{x^2} dx = -du$$

$$(4) \int \frac{dx}{1+e^{-2x}}$$

$$= \int \frac{dx}{1+\frac{1}{e^{2x}}}$$

$$= \int \frac{e^{2x}}{e^{2x}+1} dx$$

$$\text{Let } 1+e^{2x} = u$$

$$\Rightarrow 2e^{2x} dx = du$$

$$\Rightarrow e^{2x} dx = \frac{1}{2} du$$

$$= \frac{1}{2} \int \frac{du}{u}$$

$$= \frac{1}{2} \ln|u| + c$$

$$= \frac{1}{2} \ln|1+e^{2x}| + c$$

$$(5) \int \frac{1}{1-\cos x} dx$$

$$= \int \frac{1}{2 \sin^2 \frac{x}{2}} dx$$

$$= \frac{1}{2} \int \operatorname{cosec}^2 \frac{x}{2} dx$$

$$= \frac{1}{2} \left(\frac{-\cot \frac{x}{2}}{\frac{x}{2}} \right) + c$$

$$= -\cot \frac{x}{2} + c$$

$$(6) \int \frac{dx}{a^2+x^2}, \quad a \neq 0 \text{ is a constant.}$$

$$= \int \frac{a \left(\frac{dx}{a} \right)}{a^2 \left(1 + \left(\frac{x}{a} \right)^2 \right)}$$

$$= \frac{1}{a} \int \frac{dx/a}{1 + \left(\frac{x}{a} \right)^2}$$

$$= \frac{1}{a} \int \frac{du}{1+u^2}$$

$$= \frac{1}{a} \tan^{-1} u + c$$

$$= \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + c$$

$$\int \frac{du}{a^2+u^2} = \frac{1}{a} \tan^{-1} \frac{u}{a} + c$$

$$\int \frac{du}{\sqrt{a^2-u^2}} = \sin^{-1} \left(\frac{u}{a} \right) + c$$

$$\int \frac{du}{u \sqrt{u^2-a^2}} = \frac{1}{a} \sec^{-1} \left(\frac{u}{a} \right) + c$$

Practice :

5.3 → All examples;

1-12, 15-55, 61-62, 69, 71, 75

$$\text{Let, } u = \frac{x}{a}$$

$$\Rightarrow du = \frac{1}{a} dx$$

Integration by parts:

Q1

$$\int (uv) dx = u \int v dx - \int \left\{ \frac{d}{dx} u \int v dx \right\} dx$$

Usually, we choose the u -function according to the following rule,

L I A T E

L = logarithmic
I = inverse trigonometric
A = algebraic
T = trigonometric

when there are trigonometric and exponential functions, then there is no rule.

E = exponential.

Example (1) $\int x \tan^2 x dx$

$$= x \int \tan^2 x dx - \int \left\{ \frac{d}{dx} (x) \int \tan^2 x dx \right\} dx$$

$$= x \int (\sec^2 x - 1) dx - \int \left\{ (\sec^2 x - 1) dx \right\} dx$$

$$= x (\tan x - x) - \int (\tan x - x) dx$$

$$= x \tan x - x^2 - (-\ln |\cos x|) + \frac{x^2}{2} + c$$

$$= x \tan x + \ln |\cos x| + \frac{x^2}{2} + c$$

(Ans.)

$$(2) \int e^x \sin 2x dx$$

$$\text{Let } I = \int e^x \sin 2x dx$$

$$= e^x \int \sin 2x dx - \int \left\{ \frac{d}{dx}(e^x) \int \sin 2x dx \right\} dx$$

$$= e^x \left(\frac{-\cos 2x}{2} \right) - \int e^x \left(-\frac{\cos 2x}{2} \right) dx$$

$$= -\frac{1}{2} e^x \cos 2x + \frac{1}{2} \int e^x \cos 2x dx$$

$$= -\frac{1}{2} e^x \cos 2x + \frac{1}{2} \left[e^x \int \cos 2x dx - \int \left\{ \frac{d}{dx}(e^x) \int \cos 2x dx \right\} dx \right]$$

$$= -\frac{1}{2} e^x \cos 2x + \frac{1}{2} \left[e^x \cdot \frac{\sin 2x}{2} - \int e^x \cdot \frac{\sin 2x}{2} dx \right]$$

$$= -\frac{1}{2} e^x \cos 2x + \frac{1}{4} e^x \sin 2x - \frac{1}{4} \int e^x \sin 2x dx$$

$$= -\frac{1}{2} e^x \cos 2x + \frac{1}{4} e^x \sin 2x - \frac{1}{4} I$$

$$\Rightarrow I + \frac{1}{4} I = -\frac{1}{2} e^x \cos 2x + \frac{1}{4} e^x \sin 2x$$

$$\Rightarrow \frac{5}{4} I = -\frac{1}{2} e^x \cos 2x + \frac{1}{4} e^x \sin 2x$$

$$\Rightarrow I = -\frac{2}{5} e^x \cos 2x + \frac{1}{5} e^x \sin 2x + c$$

(Ans.)

Tabular method for repeated integral by parts:

Integrals of the form $\int p(x) f(x) dx$, where $p(x)$ is a polynomial can sometimes be evaluated using repeated integration by parts. Here u is taken to be $p(x)$ and repeated differentiation of $p(x)$ will eventually produce 0.

1. $\int (x^3 + x^2 - x) \cos x dx$

Repeated differentiation

$$x^3 + x^2 - x$$

$$3x^2 + 2x - 1$$

$$6x + 2$$

$$6$$

$$0$$

Repeated integration

$$\cos x$$

$$+\sin x$$

$$-\cos x$$

$$-\sin x$$

$$\cos x$$

$$\begin{aligned} \int (x^3 + x^2 - x) \cos x dx &= (x^3 + x^2 - x) \sin x + (3x^2 + 2x - 1) \cos x \\ &\quad - (6x + 2) \sin x - 6 \cos x + c \end{aligned}$$

$$= (x^3 + x^2 - x - 6x - 2) \sin x + (3x^2 + 2x - 1 - 6) \cos x + c$$

$$= (x^3 + x^2 - 7x - 2) \sin x + (3x^2 + 2x - 7) \cos x + c$$

(Ans)

(9)

Reduction formula;

Q.1 Obtain the reduction formula for $\int \sin^n x dx$. Hence evaluate $\int \sin^6 x dx$.

Solⁿ: Let $I_n = \int \sin^n x dx$

$$= \int \sin^{n-1} x \sin x dx$$

$$= \sin^{n-1} x \int \sin x dx - \int \left\{ \frac{d}{dx} (\sin^{n-1} x) \int \sin x dx \right\} dx$$

$$= -\sin^{n-1} x \cos x + \int (n-1) \sin^{n-2} x (+\cos x) dx (-\cos x) dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \cos^2 x dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int (\sin^{n-2} x - \sin^n x) dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x dx - (n-1) \int \sin^n x dx$$

$$\Rightarrow I_n = -\sin^{n-1} x \cos x + (n-1) I_{n-2} - (n-1) I_n$$

$$\Rightarrow I_n + (n-1) I_n = -\sin^{n-1} x \cos x + (n-1) I_{n-2}, \quad n \geq 2$$

$$\Rightarrow I_n = \frac{-\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}, \quad n \geq 2$$

$$\text{Now, } I_6 = \int \sin^6 x dx$$

$$= -\frac{\sin^5 x \cos x}{6} + \frac{5}{6} I_4$$

$$= -\frac{1}{6} \sin^5 x \cos x + \frac{5}{24} \left[-\frac{\sin^3 x \cos x}{4} + \frac{3}{4} I_2 \right]$$

$$= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x + \frac{5}{8} \left[-\frac{1}{2} \sin x \cos x + \frac{1}{2} I_0 \right]$$

$$= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x - \frac{5}{16} \sin x \cos x + \frac{5}{16} \int dx$$

$$= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x - \frac{5}{16} \sin x \cos x + \frac{5}{16} x + c.$$

(Ans)

Q2/ Obtain the reduction formula for $\int \sec^n x dx$ and hence evaluate $\int \sec^5 x dx$.

Solⁿ: Let $I_n = \int \sec^n x dx$

$$= \int \sec^{n-2} x \sec^2 x dx$$

$$= \sec^{n-2} x \int \sec^2 x dx - \int \left\{ \frac{d}{dx} (\sec^{n-2} x) \int \sec^2 x dx \right\} dx$$

$$= \sec^{n-2} x \tan x - \int (n-2) \sec^{n-3} x \sec x \tan x (\tan x) dx$$

$$= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-3} x \sec x \tan^2 x dx$$

$$= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x (1 - \sec^2 x) dx \quad (\sec^2 x - 1) dx$$

$$= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x (-\sec^2 x) dx$$

$$= \sec^{n-2} x \tan x - (n-2) \int \sec^n x dx + (n-2) \int \sec^{n-2} x dx$$

$$= \sec^{n-2} x \tan x - (n-2) I_n + (n-2) I_{n-2}$$

$$\Rightarrow I_n + (n-2)I_n = \sec^{n-2} x \tan x + (n-2)I_{n-2}$$

$$\Rightarrow I_n = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}$$

$$\text{Now, } I_5 = \frac{\sec^3 x \tan x}{4} + \frac{3}{4} I_3$$

$$= \frac{\sec^3 x \tan x}{4} + \frac{3}{4} \left[\frac{1}{2} \sec x \tan x + \frac{1}{2} I_1 \right]$$

$$= \frac{1}{4} \sec^3 x \tan x + \frac{3}{8} \sec x \tan x + \frac{3}{8} \int \sec x \, dx$$

$$= \frac{1}{4} \sec^3 x \tan x + \frac{3}{8} \sec x \tan x + \frac{3}{8} \ln |\sec x + \tan x| + c.$$

(Ans)

Homework: 1) Obtain the reduction formula for

$$\int \cos^n x \, dx \text{ and hence evaluate } \int \cos^5 x \, dx.$$

$$\int \tan^n x \, dx$$

Trigonometric integral:

Integrating products of sines and cosines,

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2} (1 + \cos 2x)$$

$\int \sin^m x \cos^n x \, dx$, where m, n are integers:

1) If m or n is odd, then split the odd powered

function using $\sin^2 x = 1 - \cos^2 x$ and $\sin x$.

or, $\cos^2 x = 1 - \sin^2 x$ and $\cos x$.

(2) If m and n are odd, split either of them using the above formula.

(3) If m and n are even, then use $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ and $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$ and then use reduction formula.

e.g (1) $\int \sin^3 x \cos^4 x \, dx$

$$= \int \sin^2 x \sin x \cos^4 x \, dx$$

$$= \int (1 - \cos^2 x) \cos^4 x \sin x \, dx$$

$$= -\int (1 - u^2) u^4 \, du$$

$$= -\int (u^4 - u^6) \, du$$

$$= \frac{u^7}{7} - \frac{u^5}{5} + c$$

$$= \frac{\cos^7 x}{7} - \frac{\cos^5 x}{5} + c.$$

$$\text{Let } \cos x = u$$

$$\Rightarrow -\sin x \, dx = du$$

$$\Rightarrow \sin x \, dx = -du$$

$$(2) \int \sin^6 x \cos^6 x dx$$

$$= \int \left\{ \frac{1}{2}(1 - \cos 2x) \right\}^3 \left\{ \frac{1}{2}(1 + \cos 2x) \right\}^3 dx$$

$$= \frac{1}{64} \int \{ (1 + \cos 2x)(1 - \cos 2x) \}^3 dx$$

$$= \frac{1}{64} \int (1 - \cos^2 2x)^3 dx$$

$$= \frac{1}{64} \int (\sin^2 2x)^3 dx$$

$$= \frac{1}{64} \int \sin^6 2x dx$$

$$\text{Let } 2x = u$$

$$\Rightarrow 2 dx = du$$

$$= \frac{1}{64} \int \frac{1}{2} \sin^6 u du$$

$$\Rightarrow dx = \frac{1}{2} du$$

$$= \frac{1}{128} \int \sin^6 u du$$

Now use reduction formula.

$$\# \int \sin mx \cos nx dx, \int \sin mx \sin nx dx, \\ \int \cos mx \cos nx dx \quad \text{where, } m, n \text{ are not equal.}$$

$$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

$$\sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha + \beta) + \cos(\alpha - \beta)]$$

e.g (1) $\int \sin 4x \sin 9x dx = \frac{1}{2} \int [\cos(4x-9x) - \cos(4x+9x)] dx$

$$= \frac{1}{2} \int (\cos 5x - \cos 13x) dx$$

$$= \frac{1}{2} \left[\frac{\sin 5x}{5} - \frac{\sin 13x}{13} \right] + c$$

$$= \frac{\sin 5x}{10} - \frac{\sin 13x}{26} + c$$

$\int \tan^m x \sec^n x dx$

1. If the power of $\sec x$ is even, then split of $\sec^2 x$ and $\sec^{n-2} x$.

$$\int \tan^3 x \sec^4 x dx = \int \tan^3 x \cdot \sec^2 x \cdot \sec^2 x dx$$

$$= \int \tan^3 x (1 + \tan^2 x) \sec^2 x dx$$

Let $u = \tan x$

$\Rightarrow du = \sec^2 x dx$

$$= \int u^3 (1 + u^2) du$$

$$= \int (u^3 + u^5) du$$

$$= \left[\frac{u^4}{4} + \frac{u^6}{6} \right] + c$$

$$= \frac{1}{4} \tan^4 x + \frac{1}{6} \tan^6 x + c$$

(Ans.)

2. If the power of $\tan x$ and $\sec x$ are odd, then split of $\sec x \tan x$.

e.g. $\int \tan^5 x \sec^3 x dx = \int \tan^4 x \sec^2 x \cdot \sec x \cdot \tan x dx$

$$= \int (\sec^2 x - 1)^2 \sec^2 x \cdot \sec x \tan x dx$$

Let $\sec x = u$

$\Rightarrow \sec x \tan x dx = du$

$$= \int (u^2 - 1)^2 \cdot u^2 du$$

$$= \int (u^4 - 2u^2 + 1) u^2 du$$

$$= \int (u^6 - 2u^4 + u^2) du$$

$$= \frac{u^7}{7} - \frac{2}{5} u^5 + \frac{1}{3} u^3 + c = \frac{1}{7} \sec^7 x - \frac{2}{5} \sec^5 x + \frac{1}{3} \sec^3 x + c.$$

(3) If the power of $\tan x$ is even (and $\sec x$ is odd), then express in $\sec x$ and then apply reduction formula.

e.g. $\int \tan^4 x \sec x dx$

$$= \int (\sec^2 x - 1)^2 \sec x dx$$

$$= \int (\sec^4 x - 2\sec^2 x + 1) \sec x dx$$

$$= \int (\sec^5 x - 2\sec^3 x + \sec x) dx$$

Now apply reduction formula.

Trigonometric substitution:

(2)

$$\sqrt{a^2 + x^2}$$

$$a\sqrt{x^2 - a^2}$$

$$\sqrt{a^2 - x^2}$$

$$x = a \tan \theta$$

$$x = a \sec \theta$$

$$x = a \cos \theta / a \sin \theta$$

$$\sqrt{a^2 + a^2 \tan^2 \theta}$$

$$\sqrt{a^2 \sec^2 \theta - a^2}$$

$$\sqrt{a^2 - a^2 \cos^2 \theta}$$

$$= \sqrt{a^2 \sec^2 \theta}$$

$$= \sqrt{a^2 (\sec^2 \theta - 1)}$$

$$= \sqrt{a^2 (1 - \cos^2 \theta)}$$

$$= a \sec \theta$$

$$= \sqrt{a^2 \tan^2 \theta}$$

$$= \sqrt{a^2 \sin^2 \theta}$$

$$= a \tan \theta$$

$$= a \sin \theta$$

$$a \cos \theta$$

e.g. (1) $\int \frac{dx}{x^2 \sqrt{9-x^2}}$

Let $x = 3 \sin \theta$

$\Rightarrow dx = 3 \cos \theta d\theta$

$$= \int \frac{3 \cos \theta d\theta}{9 \sin^2 \theta \sqrt{9-9 \sin^2 \theta}}$$

$$= \int \frac{3 \cos \theta d\theta}{9 \sin^2 \theta \cdot 3 \cos \theta}$$

$$= \frac{1}{9} \int \frac{d\theta}{\sin^2 \theta}$$

$$= \frac{1}{9} \int \operatorname{cosec}^2 \theta d\theta$$

$$= -\frac{1}{9} \cot \theta + c$$

$$= -\frac{1}{9} \frac{\sqrt{9-x^2}}{x} + c$$

(Ans.)

$$\sin \theta = \frac{x}{3}$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta}$$

$$= \sqrt{1 - \frac{x^2}{9}}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$= \frac{\sqrt{1 - \frac{x^2}{9}}}{x/3}$$

$$= \frac{\sqrt{9-x^2}}{x}$$

(2) ~~$5^{3/2} \sin^3 \theta \sqrt{5-5\sin^2 \theta}$~~

$$\int x^3 \sqrt{5-x^2} dx$$

$$\text{Let } x = \sqrt{5} \sin \theta$$

$$\Rightarrow dx = \sqrt{5} \cos \theta d\theta$$

$$= \int 5^{3/2} \sin^3 \theta \sqrt{5-5\sin^2 \theta} \cdot \sqrt{5} \cos \theta d\theta$$

$$= 5^{5/2} \int \sin^3 \theta \cdot \sqrt{5} \cos \theta \cdot \sqrt{5} \cos \theta d\theta$$

$$= 5^{5/2} \int \sin^3 \theta \cdot \cos^2 \theta d\theta$$

$$= 5^{5/2} \int \sin^2 \theta \cdot \sin \theta \cdot \cos^2 \theta d\theta$$

$$= 5^{5/2} \int (1 - \cos^2 \theta) \cos^2 \theta \sin \theta d\theta$$

$$\text{Let } \cos \theta = u$$

$$\Rightarrow -\sin \theta d\theta = du$$

$$\Rightarrow \sin \theta d\theta = -du$$

$$= 5^{5/2} \int (1-u^2) u^2 (-du)$$

$$= 5^{5/2} \int (u^4 - u^2) du$$

$$= 5^{5/2} \left(\frac{u^5}{5} - \frac{u^3}{3} \right) + c$$

$$\sin \theta = \frac{x}{\sqrt{5}}$$

$$= 5^{5/2} \left(\frac{\cos^5 \theta}{5} - \frac{\cos^3 \theta}{3} \right) + c$$

$$\therefore \cos \theta = \sqrt{1 - \frac{x^2}{5}}$$

$$= 5^{5/2} \left[\frac{1}{5} \left(\sqrt{1 - \frac{x^2}{5}} \right)^5 - \frac{1}{3} \left(\sqrt{1 - \frac{x^2}{5}} \right)^3 \right] + c$$

$$= 5^{5/2} \left[\frac{1}{5} \left(1 - \frac{x^2}{5} \right)^{5/2} - \frac{1}{3} \left(1 - \frac{x^2}{5} \right)^{3/2} \right] + c$$

(Ans.)

$$(3) \int \frac{dx}{x^2 \sqrt{x^2+25}}$$

$$\text{Let } x = 5 \tan \theta$$

$$\Rightarrow dx = 5 \sec^2 \theta d\theta$$

$$= \int \frac{5 \sec^2 \theta d\theta}{5^2 \tan^2 \theta \sqrt{25 \tan^2 \theta + 25}}$$

$$= \int \frac{5 \sec^2 \theta d\theta}{5^2 \tan^2 \theta \cdot 5 \sec \theta}$$

$$= \frac{1}{25} \int \frac{\sec \theta}{\tan^2 \theta} d\theta$$

$$= \frac{1}{25} \int \left(\frac{1}{\cos \theta} \times \frac{\cos^2 \theta}{\sin^2 \theta} \right) d\theta$$

$$= \frac{1}{25} \int \frac{\cos \theta}{\sin^2 \theta} d\theta$$

$$\text{Let } \sin \theta = u$$

$$\Rightarrow \cos \theta d\theta = du$$

$$= \frac{1}{25} \int \frac{1}{u^2} du$$

$$= \frac{1}{25} \left[-\frac{1}{u} \right] + c = -\frac{1}{25} \frac{1}{\sin \theta} + c$$

$$= -\frac{1}{25} \frac{\sqrt{x^2+25}}{x} + c$$

(Ans.)

$$\left. \begin{array}{l} \tan \theta = \frac{x}{5} \\ \sin \theta = \frac{x}{\sqrt{x^2+25}} \end{array} \right\}$$

$$(4) \int \frac{3x^3 dx}{\sqrt{x^2-25}}$$

$$\text{Let } x = 5 \sec \theta$$

$$\Rightarrow dx = 5 \sec \theta \tan \theta d\theta$$

$$= \int \frac{3 \times 125 \cdot \sec^3 \theta \cdot 5 \sec \theta \tan \theta d\theta}{\sqrt{5^2 \sec^2 \theta - 25}}$$

$$= \int \frac{15 \times 125 \sec^4 \theta \tan \theta d\theta}{5 \tan \theta}$$

$$= 375 \int \sec^4 \theta d\theta = 375 \int \sec^2 \theta \cdot \sec^2 \theta d\theta$$

$$= 375 \int (1 + \tan^2 \theta) \sec^2 \theta d\theta$$

$$\left| \begin{array}{l} \text{Let } \tan \theta = u \\ \Rightarrow \sec^2 \theta d\theta = du \end{array} \right.$$

$$= 375 \int (1 + u^2) du$$

$$= 375 \left[u + \frac{u^3}{3} \right] + c$$

$$\sec \theta = \frac{x}{5}$$

$$\tan \theta = \sqrt{\frac{x^2}{25} - 1}$$

$$= \frac{\sqrt{x^2 - 25}}{5}$$

$$= 375 \left[\tan \theta + \frac{\tan^3 \theta}{3} \right] + c$$

$$= 375 \cdot \frac{\sqrt{x^2 - 25}}{5} + 375 \cdot \frac{1}{3} \left(\frac{\sqrt{x^2 - 25}}{5} \right)^3 + c$$

$$= 75 \sqrt{x^2 - 25} + \left(\sqrt{x^2 - 25} \right)^3 + c.$$

▣ Integrals involving $ax^2 + bx + c$:

We need to complete the square first.

e.g (1) $\int \frac{x dx}{x^2 + 2x + 2}$

$$= \int \frac{x dx}{(x+1)^2 + 1}$$

$$\text{Let } x+1 = u$$

$$\Rightarrow dx = du$$

$$= \int \frac{(u-1) du}{u^2 + 1}$$

$$= \frac{1}{2} \int \frac{2u}{u^2 + 1} du - \int \frac{du}{u^2 + 1}$$

$$= \frac{1}{2} \ln |u^2 + 1| - \tan^{-1} u + c$$

$$= \frac{1}{2} \ln |(x+1)^2 + 1| - \tan^{-1} (x+1) + c$$

$$= \frac{1}{2} \ln |x^2 + 2x + 1 + 1| - \tan^{-1}(x+1) + C$$

$$= \frac{1}{2} \ln |x^2 + 2x + 2| - \tan^{-1}(x+1) + C.$$

(Ans.)

□ Integrating rational functions by partial fractions:

Partial fraction decomposition of $\frac{f(x)}{g(x)}$

1. If the degree of $f(x)$ is not lower than the degree of $g(x)$ then apply long division to obtain the proper form.
2. Factorise $g(x)$ using factors $ax+b$ or irreducible quadratic factors ax^2+bx+c .
3. i) For each factor $(ax+b)^n$ with $n \geq 1$, the partial fraction decomposition contains a sum of n partial fractions of the form

$$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \dots + \frac{A_n}{(ax+b)^n}$$

where each number A_k is a real number.

- ii) For $(ax^2+bx+c)^n$ with $n \geq 1$ the decomposition is,

$$\frac{A_1x+B_1}{ax^2+bx+c} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + \dots + \frac{A_nx+B_n}{(ax^2+bx+c)^n}$$

e.g (1) $\int \frac{dx}{x^3-x^2} = \int \frac{dx}{x^2(x-1)}$

Solⁿ: Let $\frac{1}{x^2(x-1)} \equiv \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1}$

$$\Rightarrow 1 \equiv Ax(x-1) + B(x-1) + Cx^2 \quad \text{--- (1)}$$

putting $x=0 \Rightarrow B=-1$

$x=1 \Rightarrow C=1$

equating the coefficients x^2 from both side of (1) $\rightarrow$

$$0 = A + C$$

$$\Rightarrow A = -C$$

$$\Rightarrow A = -1$$

$$\therefore \frac{1}{x^2(x-1)} = -\frac{1}{x} - \frac{1}{x^2} + \frac{1}{x-1}$$

$$\therefore \int \frac{dx}{x^3-x^2} = -\int \frac{dx}{x} - \int \frac{dx}{x^2} + \int \frac{dx}{x-1}$$

$$= -\ln|x| + \ln\left|\frac{1}{x}\right| + \ln|x-1| + C$$

$$= \ln\left|\frac{x-1}{x}\right| + \frac{1}{x} + C.$$

(2) $\int \frac{5x^3-3x^2+7x-3}{(x^2+1)^2} dx$

Solⁿ: Let $\frac{5x^3-3x^2+7x-3}{(x^2+1)^2} \equiv \frac{Ax+B}{x^2+1} + \frac{Cx+D}{(x^2+1)^2}$

$$\Rightarrow 5x^3-3x^2+7x-3 \equiv (Ax+B)(x^2+1) + Cx+D \quad \text{--- (1)}$$

equating the like powers of x from both sides,

$$x^3: 5 = A \Rightarrow A = 5$$

$$x^2: -3 = B \Rightarrow B = -3$$

$$x: 7 = A + C \Rightarrow C = 7 - 5 = 2$$

$$x^0: -3 = B + D \Rightarrow D = -3 + 3 = 0$$

So we obtain,

$$\begin{aligned} \int \frac{5x^3 - 3x^2 + 7x - 3}{(x^2 + 1)^2} dx &= \int \frac{5x - 3}{(x^2 + 1)} dx + \int \frac{2x}{(x^2 + 1)^2} dx \\ &= \int \frac{5x dx}{x^2 + 1} - \int \frac{3 dx}{x^2 + 1} + \int \frac{2x}{(x^2 + 1)^2} dx \end{aligned}$$

let $u = x^2 + 1$
 $\Rightarrow du = 2x dx$

$$= \frac{5}{2} \int \frac{2x dx}{x^2 + 1} - 3 \int \frac{dx}{x^2 + 1} + \int \frac{du}{u^2}$$

$$= \frac{5}{2} \ln |x^2 + 1| - 3 \tan^{-1} x - \frac{1}{u} + C$$

$$= \frac{5}{2} \ln |x^2 + 1| - 3 \tan^{-1} x - \frac{1}{x^2 + 1} + C.$$

(Ans)

Exercise: 7.4 $\rightarrow$ 1-26, 37, 48

7.5 $\rightarrow$ 1-33 (odd), 39, 41.

Improper integrals:

Consider the definite integral $\int_a^b f(x) dx$.

If the integrand f has points of discontinuity on $[a, b]$ or any or both a and b are infinite, then the integral is called improper integral.

e.g. $\int_1^{\infty} \frac{dx}{x^2}$; $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$, $\int_{-\infty}^1 e^x dx$

These are improper integrals with infinite intervals.

$$\int_{-1}^1 \frac{dx}{x^2} , \int_1^2 \frac{dx}{x-1}$$

These are improper integral with infinite discontinuity in the interval of integration.

$$\int_0^{\infty} \frac{dx}{\sqrt{x}} \rightarrow \text{infinite discontinuity and infinite interval.}$$

We evaluate improper integral as,

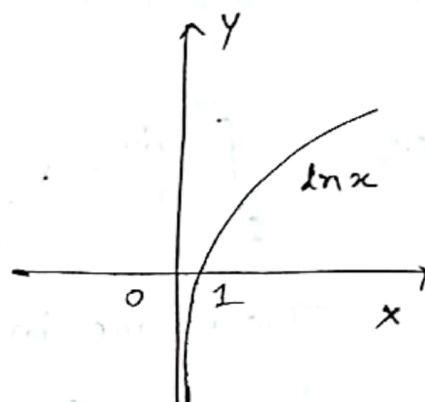
$$\int_a^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$$

$$\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx$$

If the limit exists, the improper integral is said

to converge and the limit is defined to be the value of the integral. If the limit doesn't exist, the improper integral is said to diverge.

e.g 1.
$$\int_1^{\infty} \frac{dx}{x} = \lim_{b \rightarrow \infty} \int_1^b \frac{dx}{x}$$
$$= \lim_{b \rightarrow \infty} [\ln x]_1^b$$
$$= \lim_{b \rightarrow \infty} [\ln b - \ln 1]$$
$$= \lim_{b \rightarrow \infty} \ln b$$
$$= \infty$$



The integral diverges and it has no value.

2)
$$\int_2^{\infty} \frac{dx}{(x-1)^2} = \lim_{b \rightarrow \infty} \int_2^b \frac{dx}{(x-1)^2}$$
$$= \lim_{b \rightarrow \infty} \left[-\frac{1}{x-1} \right]_2^b$$
$$= \lim_{b \rightarrow \infty} \left[-\frac{1}{b-1} + \frac{1}{1} \right]$$
$$= \lim_{b \rightarrow \infty} \left[1 - \frac{1}{b-1} \right]$$
$$= 1$$

Thus the integral converges and the value is 1.

(17)

Que (3) $\int_0^{\infty} (1-x)e^{-x} dx = \lim_{b \rightarrow \infty} \int_0^b (1-x)e^{-x} dx$

Now, $\int_0^b (1-x)e^{-x} dx = (1-x) \int e^{-x} dx - \int \left\{ \frac{d}{dx} (1-x) \int e^{-x} dx \right\} dx$

$$= -(1-x)e^{-x} - \int e^{-x} dx$$

$$= -(1-x)e^{-x} + e^{-x} + c$$

$$= (x-1)e^{-x} + e^{-x} + c$$

$$= e^{-x} (x-1+1) + c$$

$$= x e^{-x} + c$$

$$\lim_{b \rightarrow \infty} \int_0^b (1-x)e^{-x} dx = \lim_{b \rightarrow \infty} [x e^{-x}]_0^b$$

$$= \lim_{b \rightarrow \infty} b e^{-b} ; \infty \times 0 \text{ form}$$

$$= \lim_{b \rightarrow \infty} \frac{b}{e^b} ; \frac{\infty}{\infty} \text{ form}$$

$$= \lim_{b \rightarrow \infty} \frac{1}{e^b} \quad [\text{Applying L' Hopital rule}]$$

$$= 0$$

The integral converges and it has value 0.

(4) $\int_{-\infty}^1 e^x dx = \lim_{a \rightarrow -\infty} \int_a^1 e^x dx = \lim_{a \rightarrow -\infty} [e^x]_a^1$

$$= \lim_{a \rightarrow -\infty} [e - e^a]$$

$$= e$$

The improper integral of f over the interval $(-\infty, \infty)$ is defined as,

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^{\infty} f(x) dx$$

where c is any real number. The improper integral is said to converge if both term converge and diverge if both term either of them diverge.

e.g. $\int_{-\infty}^{\infty} \frac{e^{-t}}{1+e^{-2t}} dt$

Let $e^{-t} = u$

$$\Rightarrow -e^{-t} dt = du$$

$$\Rightarrow e^{-t} dt = -du$$

Now, $\int \frac{-du}{1+u^2}$

$$= -\tan^{-1} u + c$$

$$= -\tan^{-1}(e^{-t}) + c.$$

$$\text{So, } \int_{-\infty}^{\infty} \frac{e^{-t}}{1+e^{-2t}} dt = \int_{-\infty}^0 \frac{e^{-t} dt}{1+e^{-2t}} + \int_0^{\infty} \frac{e^{-t} dt}{1+e^{-2t}}$$

$$= \lim_{a \rightarrow -\infty} \int_a^0 \frac{e^{-t} dt}{1+e^{-2t}} + \lim_{b \rightarrow \infty} \int_0^b \frac{e^{-t} dt}{1+e^{-2t}}$$

$$= \lim_{a \rightarrow -\infty} [-\tan^{-1}(e^{-t})]_a^0 + \lim_{b \rightarrow \infty} [-\tan^{-1}(e^{-t})]_0^b$$

$$= \lim_{a \rightarrow -\infty} [-\tan^{-1}(1) + \tan^{-1}(e^{-a})] + \lim_{b \rightarrow \infty} [-\tan^{-1}(e^{-b}) + \tan^{-1}(1)]$$

$$= -\frac{\pi}{4} + \tan^{-1}(\infty) - \tan^{-1}(0) + \frac{\pi}{4} = \frac{\pi}{2} \text{ (Ans.)}$$

e.g. Evaluate, $\int_1^2 \frac{dx}{1-x}$

ke Now, $\lim_{b \rightarrow 1^+} \int_b^2 \frac{dx}{1-x}$

$$= \lim_{b \rightarrow 1^+} \left[-\ln|1-x| \right]_b^2$$

$$= \lim_{b \rightarrow 1^+} \left[-\ln|-1| + \ln|1-b| \right]$$

$$= \lim_{b \rightarrow 1^+} \ln|1-b|$$

$$= -\infty.$$

So the interval diverges.

e.g. $\int_1^4 \frac{dx}{(x-2)^{2/3}}$

The integral is improper because the integrand approaches $+\infty$ at $x=2$.

$$\text{So, } \int_1^4 \frac{dx}{(x-2)^{2/3}} = \int_1^2 \frac{dx}{(x-2)^{2/3}} + \int_2^4 \frac{dx}{(x-2)^{2/3}}$$

$$= \lim_{b \rightarrow 2} \int_1^b \frac{dx}{(x-2)^{2/3}} + \lim_{a \rightarrow 2} \int_a^4 \frac{dx}{(x-2)^{2/3}}$$

$$= \lim_{b \rightarrow 2} \left[3(x-2)^{1/3} \right]_1^b + \lim_{a \rightarrow 2} \left[3(x-2)^{1/3} \right]_a^4$$

$$= \lim_{b \rightarrow 1} [3(b-2)^{1/3} - 3(1-2)^{1/3}] + 3 \lim_{a \rightarrow 2} [3(4-2)^{1/3} - 3(a-2)^{1/3}]$$

$$= 3 + 3 \cdot 3\sqrt{2}$$

(Ans.)

e.g. : $\int_{-2}^2 \frac{dx}{x^2}$

The integral is improper because the integrand approaches $+\infty$ at $x=0$.

$$\text{Now, } \int_{-2}^2 \frac{dx}{x^2} = \int_{-2}^0 \frac{dx}{x^2} + \int_0^2 \frac{dx}{x^2}$$

$$= \lim_{b \rightarrow 0} \int_{-2}^b \frac{dx}{x^2} + \lim_{a \rightarrow 0} \int_a^2 \frac{dx}{x^2}$$

$$= \lim_{b \rightarrow 0} \left[-\frac{1}{x} \right]_{-2}^b + \lim_{a \rightarrow 0} \left[-\frac{1}{x} \right]_a^2$$

$$= \lim_{b \rightarrow 0} \left[-\frac{1}{b} + \frac{1}{2} \right] + \lim_{a \rightarrow 0} \left[-\frac{1}{2} + \frac{1}{a} \right]$$

$$= -\infty + +\infty$$

The integrand is divergent.

The Gamma function:

$$\Gamma n = \int_0^{\infty} e^{-x} x^{n-1} dx$$

$$\Gamma_{n+1} = n!$$

$$\Gamma n = (n-1)!$$

$$\Gamma_{1/2} = \sqrt{\pi}$$

The Beta function:

The Beta function $\beta(m, n)$ is defined by the definite integral.

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx, \quad m, n > 0 \quad \text{--- (1)}$$

$\beta(m, n)$ is symmetric, i.e. $\beta(m, n) = \beta(n, m)$.

If $x = 1-y$ then,

$$dx = -dy \quad \therefore 1-x = y$$

when $x=0$, $y=1$

$x=1$, $y=0$

$$\begin{aligned} \beta(m, n) &= \int_0^1 x^{m-1} (1-x)^{n-1} dx \\ &= \int_1^0 (1-y)^{m-1} y^{n-1} (-dy) \\ &= \int_0^1 y^{n-1} (1-y)^{m-1} dy \\ &= \beta(n, m). \end{aligned}$$

$\therefore \beta(m, n) = \beta(n, m)$ i.e. $\beta(m, n)$ is symmetric.

Q The relation between Beta function and Gamma function.

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

Show that. $\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$ [no need]

We know $\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$

Let $x = \frac{y}{1+y}$

$$\Rightarrow dx = \frac{1}{(1+y)^2} dy$$

$x=0, y=0$

$$\frac{1}{x} = \frac{1+y}{y} = 1 + \frac{1}{y}$$

If $x=1$ then $1 = 1 + \frac{1}{y}$

$$\Rightarrow \frac{1}{y} = 0$$

$$\Rightarrow y \rightarrow \infty$$

$$1-x = 1 - \frac{y}{1+y} = \frac{1}{1+y}$$

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$= \int_0^{\infty} \left(\frac{y}{1+y} \right)^{m-1} \left(\frac{1}{1+y} \right)^{n-1} \frac{dy}{(1+y)^2}$$

$$= \int_0^{\infty} \frac{y^{m-1}}{(1+y)^{m+n}} dy \quad \text{--- (1)}$$

properties:

$$1. \Gamma(m) \Gamma(1-m) = \frac{\pi}{\sin m\pi}, \quad 0 < m < 1$$

$$2. \text{ Show that, } \int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

Solution: We know that.

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \quad \text{--- (1)}$$

$$\text{Let } x = \sin^2 \theta$$

$$\Rightarrow dx = 2 \sin \theta \cos \theta d\theta$$

$$\text{When } x=0, \quad \theta=0 \quad \text{so,}$$

$$x=1, \quad \theta = \frac{\pi}{2}$$

$$\begin{aligned} \beta(m, n) &= \int_0^{\pi/2} \sin^{2m-2} \theta (1 - \sin^2 \theta)^{n-1} \cdot 2 \sin \theta \cos \theta d\theta \\ &= 2 \int_0^{\pi/2} \sin^{2m-2} \theta \cos^{2n-2} \theta \sin \theta \cos \theta d\theta \\ &= 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta \end{aligned}$$

$$\text{putting } 2m-1=p, \quad 2n-1=q$$

$$\Rightarrow m = \frac{p+1}{2} \quad \Rightarrow n = \frac{q+1}{2}$$

$$\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right) = 2 \int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta$$

$$\Rightarrow \int_0^{\pi/2} \sin^p \theta \cos^q \theta \, d\theta = \frac{1}{2} \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

$$= \frac{1}{2} \frac{\sqrt{\frac{p+1}{2}} \sqrt{\frac{q+1}{2}}}{\sqrt{\left(\frac{p+q+2}{2}\right)}}$$

e.g (1) $\int_0^{\pi/2} \sin^3 \theta \cos^4 \theta \, d\theta = \frac{1}{2} \beta\left(\frac{3+1}{2}, \frac{4+1}{2}\right)$

$$= \frac{1}{2} \beta\left(2, \frac{5}{2}\right)$$

$$= \frac{1}{2} \beta \frac{\sqrt{2} \sqrt{5/2}}{\sqrt{9/2}}$$

$$= \frac{1}{2} \frac{1 \sqrt{5/2}}{\frac{3}{2} \sqrt{7/2}} \quad \left[\because \Gamma(n+1) = n\Gamma(n) \right]$$

$$= \frac{1}{2} \frac{\sqrt{5/2}}{\frac{3}{2} \cdot \frac{5}{2} \sqrt{5/2}}$$

$$= \frac{1}{2} \frac{4}{35} = \frac{2}{35}$$

(2) $\int_0^{\pi/2} \cos^6 \theta \, d\theta = \frac{1}{2} \beta\left(\frac{0+1}{2}, \frac{6+1}{2}\right)$

$$= \frac{1}{2} \beta\left(\frac{1}{2}, \frac{7}{2}\right)$$

$$= \frac{1}{2} \frac{\sqrt{1/2} \sqrt{7/2}}{\sqrt{4}}$$

$$= \frac{1}{2} \frac{\sqrt{1/2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{1/2}}{3!}$$

$$= \frac{1}{2} \frac{\sqrt{\pi} \cdot 5 \cdot 3 \cdot \sqrt{\pi}}{6 \cdot 8} = \frac{5\pi}{32} \quad (\text{Ans})$$

$$3) \int_0^1 \frac{dx}{(1-x^3)^{1/3}}$$

$$\text{Let } x^3 = z$$

$$\Rightarrow x = z^{1/3}$$

$$\Rightarrow dx = \frac{1}{3} z^{-2/3} dz$$

$$\text{when } x=0, z=0$$

$$x=1, z=1$$

$$= \int_0^1 \frac{\frac{1}{3} z^{-2/3} dz}{(1-z)^{1/3}}$$

$$= \frac{1}{3} \int_0^1 z^{-2/3} (1-z)^{-1/3} dz$$

$$= \frac{1}{3} \int_0^1 z^{1/3-1} (1-z)^{2/3-1} dz$$

$$= \frac{1}{3} B\left(\frac{1}{3}, \frac{2}{3}\right)$$

$$= \frac{1}{3} \frac{\Gamma^{1/3} \Gamma^{2/3}}{\Gamma 1} = \frac{1}{3} \Gamma^{1/3} \Gamma^{1-1/3}$$

$$= \frac{1}{3} \frac{\pi}{\sin \pi/3} \quad \left[\Gamma m \Gamma 1-m = \frac{\pi}{\sin m\pi}; 0 < m < 1 \right]$$

$$= \frac{2\pi}{3\sqrt{3}}$$

$$4) \int_0^\infty x^{1/4} e^{-\sqrt{x}} dx$$

$$= \int_0^\infty e^{-\sqrt{x}} (\sqrt{x})^{1/2} dx$$

$$\text{Let } \sqrt{x} = z$$

$$\Rightarrow x = z^2$$

$$= \int_0^\infty e^{-z} z^{1/2} \cdot 2z dz$$

$$\Rightarrow dx = 2z dz$$

$$= 2 \int_0^\infty e^{-z} z^{3/2} dz$$

$$x=0 \quad z=0$$

$$x=\infty \quad z=\infty$$

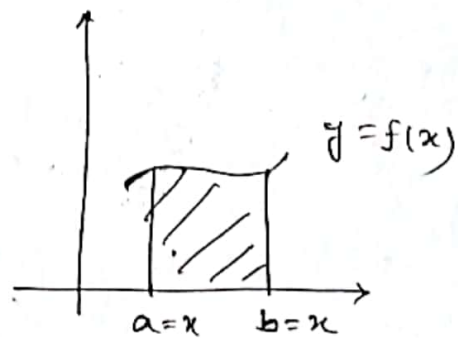
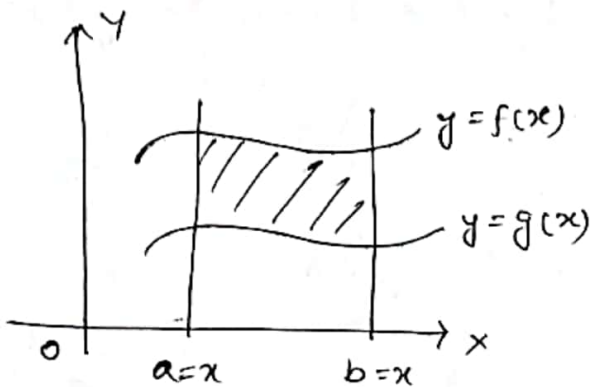
$$= 2 \int_0^\infty e^{-z} z^{5/2-1} dz = 2 \Gamma^{5/2} = 2 \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma^{1/2} = \frac{3}{2} \sqrt{\pi}$$

Application of definite integral

Area between two curves:

If f and g are continuous on $[a, b]$, $f(x) \geq g(x)$ for all $x \in [a, b]$ then the area of the region bounded above by $y = f(x)$, below by $y = g(x)$, on the left by the line $x = a$ and on the right by the line $x = b$ is,

$$A = \int_a^b [f(x) - g(x)] dx$$



$$\int_a^b [f(x) - 0] dx$$

Example(1): Find the area bounded by the graph

$$y = 3 - x \text{ and } y = x^2 - 9.$$

Solⁿ: intersection point,

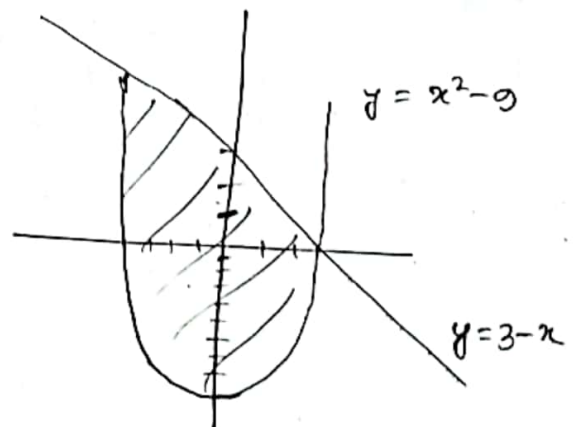
$$3 - x = x^2 - 9$$

$$\Rightarrow x^2 + x - 9 - 3 = 0$$

$$\Rightarrow x^2 + x - 12 = 0$$

$$\Rightarrow (x + 4)(x - 3) = 0$$

$$\Rightarrow x = -4, 3$$



$$A = \int_{-4}^3 (3-x-x^2+9) dx$$

$$= \int_{-4}^3 (-x^2-x+12) dx$$

$$= \left[-\frac{x^3}{3} - \frac{x^2}{2} + 12x \right]_{-4}^3$$

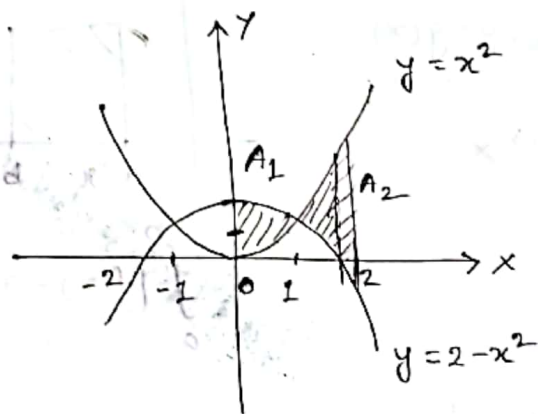
$$= \left(-\frac{27}{3} - \frac{9}{2} + 36 \right) - \left(\frac{64}{3} - \frac{16}{2} - 48 \right)$$

$$= -\frac{91}{3} + \frac{7}{2} + 84$$

$$= \frac{343}{6}$$

(2) Find the area bounded by the parabola $y=x^2$ and $y=2-x^2$ for $0 \leq x \leq 2$.

Solution:



The intersection points are,

$$x^2 = 2 - x^2$$

$$\Rightarrow 2x^2 = 2$$

$$\Rightarrow x^2 = 1$$

$$\Rightarrow x = 1, -1$$

$$A = A_1 + A_2$$

$$= \int_0^1 (2 - x^2 - x^2) dx + \int_1^2 (x^2 - 2 + x^2)$$

$$= \int_0^1 (2 - 2x^2) dx + \int_1^2 (2x^2 - 2) dx$$

$$= \left[2x - \frac{2x^3}{3} \right]_0^1 + \left[\frac{2x^3}{3} - 2x \right]_1^2$$

$$= 2 - \frac{2}{3} + \frac{16}{3} - 4 - \frac{2}{3} + 2$$

$$= \frac{16}{3} - \frac{4}{3} = \frac{12}{3} = 4.$$

Q1 Find the area between the curves $x = y^2$ and $x = 6 - y^2$.

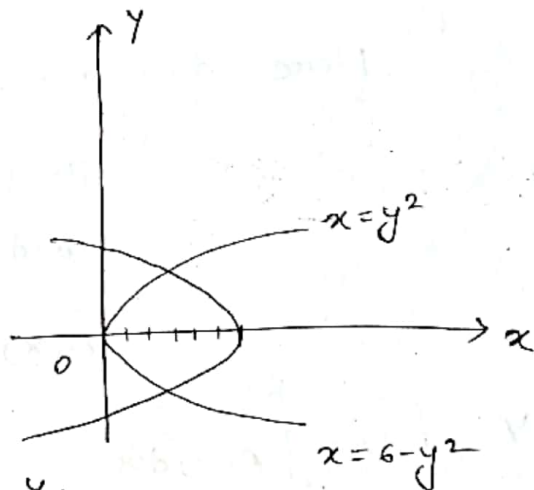
Solution: Intersecting point,

$$y^2 = 6 - y^2$$

$$\Rightarrow 2y^2 = 6$$

$$\Rightarrow y^2 = 3$$

$$\Rightarrow y = \pm \sqrt{3}$$



Integrating with respect to y .

$$A = \int_{-\sqrt{3}}^{\sqrt{3}} [f(y) - g(y)] dy$$

$$= \int_{-\sqrt{3}}^{\sqrt{3}} (6 - y^2 - y^2) dy$$

$$= \int_{-\sqrt{3}}^{\sqrt{3}} (6 - 2y^2) dy$$

$$= \left[6y - \frac{2}{3} y^3 \right]_{-\sqrt{3}}^{\sqrt{3}} = 6\sqrt{3} - \frac{2}{3} 3\sqrt{3} + 6\sqrt{3} - \frac{2}{3} 3\sqrt{3}$$

$$= 12\sqrt{3} - 2\sqrt{3} - 2\sqrt{3}$$

$$= 8\sqrt{3}$$

▮ Arc length:

$$(\Delta l)^2 = (\Delta x)^2 + (\Delta y)^2$$

$$\Rightarrow \left(\frac{\Delta l}{\Delta x}\right)^2 = 1 + \left(\frac{\Delta y}{\Delta x}\right)^2$$

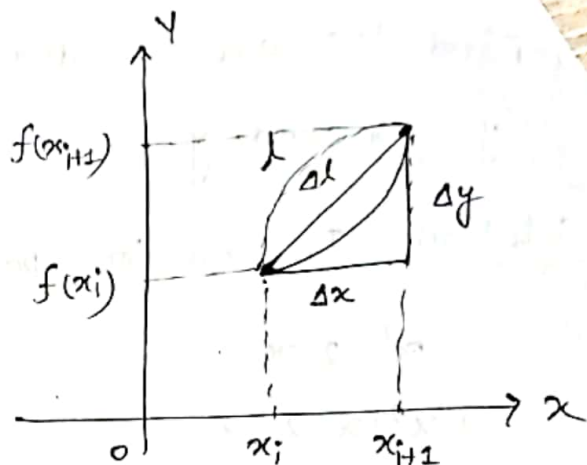
$$\lim_{\Delta x \rightarrow 0} \frac{\Delta l}{\Delta x} = \lim_{\Delta x \rightarrow 0} \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2}$$

$$\Rightarrow \frac{dl}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\therefore L = \int dl = \int \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_a^b \sqrt{1 + \{f'(x)\}^2} dx$$

Similarly, $\frac{dl}{dy} = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}$

$$\therefore L = \int dl = \int \sqrt{1 + \{f'(y)\}^2} dy$$



Example: Find the arc length at $x = \frac{1}{3}(y^2+2)^{3/2}$ from $y=0$ to $y=1$.

Solⁿ: Arc length,

$$L = \int dl = \int_0^1 \sqrt{1 + 2y^2 + y^4} dy$$

$$= \int_0^1 \sqrt{(1+y^2)^2} dy$$

$$= \int_0^1 (y^2+1) dy = \left[\frac{y^3}{3} + y\right]_0^1 = \frac{4}{3}$$

$$\frac{dx}{dy} = \frac{1}{3} \cdot \frac{3}{2} (y^2+2)^{1/2} \cdot 2y$$

$$= y(y^2+2)^{1/2}$$

$$\left(\frac{dx}{dy}\right)^2 = y^2(y^2+2)$$

$$= y^4 + 2y^2$$

Q. Arc length for parametric curve:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

example: $x = \cos t + t \sin t$
 $y = \sin t - t \cos t$, $0 \leq t < \pi$

$$\frac{dx}{dt} = -\sin t + t \cos t + \sin t = t \cos t$$

$$\frac{dy}{dt} = \cos t + t \sin t - \cos t = t \sin t$$

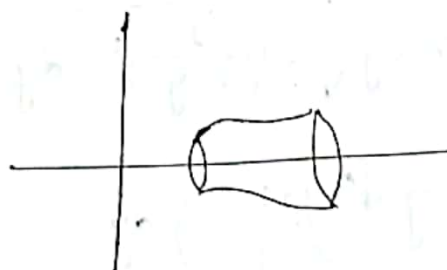
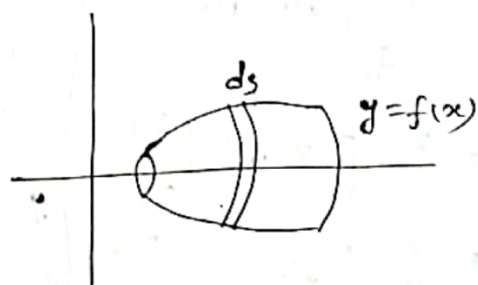
$$\therefore L = \int_0^{\pi} \sqrt{t^2 \cos^2 t + t^2 \sin^2 t} dt$$

$$= \int_0^{\pi} \sqrt{t^2 (\cos^2 t + \sin^2 t)} dt$$

$$= \int_0^{\pi} t dt = \left[\frac{t^2}{2} \right]_0^{\pi}$$

$$= \frac{\pi^2}{2}$$

Q. Area of a surface of revolution



$$dA = 2\pi r ds$$

when the curve is revolved about x-axis then,

$$dA = 2\pi r ds = 2\pi y ds = 2\pi f(x) \sqrt{1 + \{f'(x)\}^2}$$

$$\therefore A = \int_a^b 2\pi f(x) \sqrt{1 + \{f'(x)\}^2} dx$$

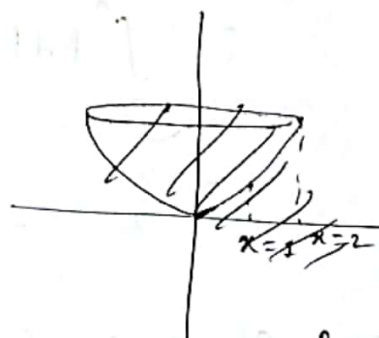
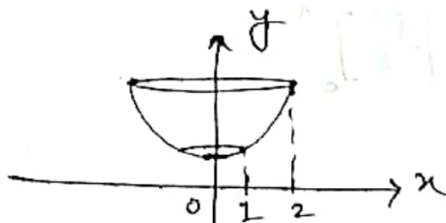
$$dA = 2\pi r ds = 2\pi x ds = 2\pi x \sqrt{1 + \{f'(y)\}^2}$$

$$\therefore A = \int_c^d 2\pi f(y) \sqrt{1 + \{f'(y)\}^2} dy$$

A surface of revolution is a surface that is generated by revolving a plane curve about x -axis that lies in the same plane as the curve.

Q1 Find the area of the surface that is generated by revolving the portion of the curve $y = x^2$ between $x=1$ and $x=2$ about y -axis.

Solⁿ:



$$A = \int 2\pi x \sqrt{1 + \{f'(y)\}^2} dy$$

$$= \int_1^4 2\pi \sqrt{y} \sqrt{1 + \left(\frac{1}{2\sqrt{y}}\right)^2} dy$$

$$= \int_1^4 \left(2\pi \sqrt{y} \frac{\sqrt{4y+1}}{2\sqrt{y}} \right) dy$$

$$= \pi \int_1^4 \sqrt{4y+1} dy$$

$$= \pi \int_5^{17} \frac{1}{4} \sqrt{u} du$$

$$= \frac{\pi}{4} \left[\frac{2}{3} u^{3/2} \right]_5^{17}$$

$$= \frac{\pi}{6} [17^{3/2} - 5^{3/2}]$$

(Ans)

$$\text{Let } 4y+1=u$$

$$\Rightarrow 4dy = du$$

$$\Rightarrow dy = \frac{1}{4} du$$

$$y=1, u=5$$

$$y=4, u=17$$